Explainable Quantum-Inspired Multi-label Classification of Non-Functional Requirements

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Abstract. Non-functional requirements (NFRs) frequently overlap across quality attributes such as security, performance, usability, and availability, which makes NFR classification a natural multi-label problem. This paper presents an explainable quantum-inspired approach for multi-label NFR classification. Requirements are represented as normalized semantic states, NFR labels as projection directions, and label scores as squared projection amplitudes; a label interference matrix models NFR co-occurrence, and token-level amplitude contributions provide interpretability. We evaluate several quantum-inspired variants on the NICE multi-label dataset against TF-IDF Logistic Regression, TF-IDF Linear SVM, a Sentence-BERT baseline, and a label-frequency baseline. Under 5-fold cross-validation, Sentence-BERT achieves the highest Macro-F1, while classical TF-IDF baselines and the hybrid quantum–SVM model perform comparably, with the hybrid model obtaining the highest Macro-F1 among non-embedding models. The Macro-F1 differences between the hybrid model and TF-IDF baselines are not statistically significant in a Wilcoxon signed-rank test. As a standalone classifier, the quantum-inspired projection is significantly weaker than the classical baselines (large effect size across cross-validation folds); fused with a linear SVM, its accuracy gain over TF-IDF baselines is small, directionally positive, and not statistically significant at this dataset size. The contribution of this work is therefore centered on interpretability and on a diagnostic account of where the quantum-inspired geometry helps and where it does not, rather than on an accuracy improvement. We further provide a deletion-based evaluation of the token-level explanations.

Keywords: Requirements Engineering · Non-Functional Requirements · Multi-label Classification · Quantum-inspired Model · Explainable AI · Software Engineering

1 Introduction

Requirements Engineering (RE) is a critical activity in software development because requirement quality strongly affects downstream architecture, implementation, testing, and maintenance decisions. Among requirement types, non-functional requirements (NFRs) are particularly difficult to identify and classify automatically. NFRs describe quality attributes such as security, performance, usability, availability, scalability, and maintainability. Unlike many functional requirements, NFRs are often ambiguous, context-dependent, and overlapping.

For example, a requirement stating that payment information must be encrypted and accessible only to authorized administrators clearly relates to security. If the same requirement also constrains availability, auditability, or response time, it may express multiple NFR categories at once. Therefore, NFR classification is better treated as a multi-label problem rather than a strict single-label problem [19,21].

This paper investigates the research topic: *Explainable Quantum-Inspired Multi-label Classification of Non-Functional Requirements*. The key idea is to represent each requirement as a semantic state and estimate its degree of association with multiple NFR labels using projection amplitudes. The approach is quantum-inspired rather than quantum-hardware-based: it runs on ordinary computers and borrows mathematical intuitions such as state representation, projection amplitude, and interference from geometric information retrieval and quantum language modeling [16,18,8].

The basic projection component is closely related to centroid-based text classification. The contribution of this work is therefore not a claim of raw performance superiority, but a combination of (i) an interpretable geometric scoring view, (ii) a label-coupling formulation via an interference term, which later experiments show is only marginally active in practice, and (iii) a hybrid calibration with a discriminative classifier, whose accuracy advantage over classical baselines we report as suggestive rather than confirmed.

The contributions of this paper are:

- a reproducible implementation for multi-label NFR classification experiments;
- quantum-inspired projection, contrastive projection, and hybrid quantum–SVM variants;
- token-level semantic amplitude explanations and a deletion-based faithfulness test;
- empirical comparison against TF-IDF Logistic Regression, TF-IDF Linear SVM, Sentence-BERT, and a label-frequency baseline on the NICE dataset;
- an ablation study of projection type, label interference, and hybrid fusion weight.

2 Related Work

Automated NFR classification has been studied for many years in Requirements Engineering. Early work by Cleland-Huang et al. explored automated classification of NFRs such as security, performance, and usability [1]. Later studies introduced supervised machine learning and dependency-based explanations for requirements classification [7,2]. The PROMISE and PROMISE-expanded datasets have supported repeatable experiments on requirement classification [11].

NICE introduced a multi-label NFR dataset based on the augmented PROMISE dataset and relabeled NFR sub-classes such as availability, security, and performance using a multi-label schema [13,14]. This dataset is especially relevant to the present work because the research problem is explicitly multi-label. A direct numeric comparison with the NICE paper is limited by differences in label filtering, splits, and evaluation protocol, so we report our own baselines on the same downloaded data.

Traditional text classification commonly relies on bag-of-words or TF-IDF features combined with classifiers such as Logistic Regression or Support Vector Machines [17]. Transformer-based models have become strong baselines for requirements classification, including BERT-style transfer learning approaches such as NoRBERT [4,6]. Sentence-BERT provides efficient sentence embeddings for downstream classifiers [12], and is therefore included as a stronger embedding baseline in this study.

The quantum-inspired component is related to geometric information retrieval and quantum-inspired language models, where term dependencies and matching can be modeled through vector spaces, projections, and complex-valued or wave-function-inspired representations [16,18,22,8]. For explainability, this work is positioned near model explanation methods such as LIME and SHAP [15,10], and evaluates faithfulness using a deletion test inspired by rationale evaluation work such as ERASER [5].

3 Research Questions

This study is guided by three research questions:

RQ1. Can a quantum-inspired semantic projection model classify multi-label NFRs competitively compared with classical and embedding baselines?

RQ2. Does the model provide useful explainability through token-level semantic amplitude contributions?

RQ3. Is the observed behavior stable under cross-validation rather than only in a single train/test split?

4 Proposed Method

4.1 Requirement Representation

Each requirement text is first converted into a TF-IDF vector:

$$\mathbf{x}_i \in \mathbb{R}^d, \tag{1}$$

where d is the vocabulary size. The vector is normalized to form a semantic state:

$$|r_i\rangle = \frac{\mathbf{x}_i}{\|\mathbf{x}_i\|_2}. \tag{2}$$

This normalization allows the requirement to be interpreted as a direction in semantic space.

4.2 Label Projection

For each NFR label c, a label basis vector is estimated from the centroid of positive training examples:

$$|c\rangle = \frac{1}{N_c} \sum_{i:y_{ic}=1} |r_i\rangle, \tag{3}$$

followed by normalization. The raw association between requirement r_i and label c is computed as a squared projection amplitude:

$$s_c(r_i) = \left(\langle c \mid r_i \rangle\right)^2. \tag{4}$$

We call $s_c(r_i)$ an association score rather than a probability because the scores over labels are not constrained to sum to one.

4.3 Contrastive Label Projection

The positive-only centroid projection is simple and interpretable, but it does not explicitly model how a label differs from other labels. Therefore, this work also evaluates a contrastive quantum-inspired variant. For each label c, the contrastive direction is computed as:

$$|c_\Delta\rangle = \frac{\mu_c^+ - \mu_c^-}{\left\|\mu_c^+ - \mu_c^-\right\|_2}, \tag{5}$$

where μ_c^+ is the centroid of requirements that contain label c, and μ_c^- is the centroid of requirements that do not contain label c. Unlike the positive centroid vector, the contrastive direction is signed: a requirement can project in the opposite direction of a label. Squaring a negative projection would incorrectly assign a high score to evidence against the label. Therefore, the contrastive variant uses a half-wave rectified projection score:

$$s_c^\Delta(r_i) = \left(\max(0, \langle c_\Delta \mid r_i \rangle)\right)^2. \tag{6}$$

This preserves positive alignment with the label direction while suppressing opposite-direction evidence.

4.4 Interference Adjustment

To reflect label co-occurrence, the model estimates a label interference matrix from training labels. Let N_{kl} be the number of training samples containing both labels k and l, and let N_k be the number of samples containing label k. The normalized co-occurrence matrix is:

$$M_{kl} = \frac{N_{kl}}{N_k}, \tag{7}$$

with the diagonal set to 1. The implementation mixes this matrix with the identity matrix:

$$\mathbf{M}' = (1 - \lambda)\mathbf{I} + \lambda\mathbf{M}, \tag{8}$$

where λ controls interference strength. For projection scores $\mathbf{p}_i$, adjusted scores are computed as:

$$\mathbf{s}_i = \mathrm{normalize}(\mathbf{p}_i\mathbf{M}'). \tag{9}$$

For the positive-projection model, normalize$(\cdot)$ denotes row-wise min–max normalization into $[0, 1]$. For the contrastive model, the rectified projection is passed through a sigmoid calibration and the interference-adjusted scores are clipped into $[0, 1]$. The resulting scores are association scores for thresholding rather than mutually exclusive probabilities.

4.5 Hybrid Quantum–SVM Fusion

To improve stability on small and imbalanced datasets, a hybrid model combines the contrastive quantum-inspired score with a one-vs-rest Linear SVM score:

$$\mathbf{h}_i = \alpha\mathbf{q}_i + (1 - \alpha)\mathbf{v}_i, \tag{10}$$

where $\mathbf{q}_i$ is the contrastive quantum-inspired score vector, $\mathbf{v}_i$ is the calibrated SVM score vector, and α is the quantum fusion weight. In the final experiments, $\alpha = 0.30$ is used as a moderate quantum weight; the ablation study (Table 5) shows that Macro-F1 is not sensitive to the exact value in this range ($\alpha = 0.15$, 0.30, and 0.50 are within one standard deviation of each other), so $\alpha = 0.30$ was kept as a round, non-edge value rather than as a value selected for maximizing cross-validation Macro-F1.

4.6 Explainability

For a predicted label c, token-level contribution is estimated from the element-wise product between the requirement state and the label basis:

$$e_{ijc} = r_{ij}c_j. \tag{11}$$

Terms with the largest absolute contributions are reported as explanations. In the experiments, these explanations are generated by the rectified contrastive projection model. We evaluate them using a deletion test: if the highlighted terms are faithful, deleting them should reduce the model score more than deleting random nonzero terms.

4.7 Theoretical Characterization

The proposed formulation exposes two independent sources of cross-label coupling in the deployed decision rule: the interference matrix (Proposition 1) and the row-wise score normalization that the implementation always applies before thresholding (Proposition 2). Only the within-instance *ranking* of labels reduces to a cosine-similarity ranking; the thresholded binary decision does not reduce to L independent per-label centroid classifiers, even when interference is switched off.

Proposition 1 (Interference as label coupling). *With* $\mathbf{M}' = (1-\lambda)\mathbf{I} + \lambda\mathbf{M}$*, the pre-normalization score for label* l *has the form*

$$u_l \propto (1-\lambda)p_l + \lambda \sum_k p_k M_{kl}. \tag{12}$$

The second term injects score mass from co-occurring labels k*, where* $M_{kl} = N_{kl}/N_k$*. This coupling cannot be represented by a diagonal per-label centroid classifier; together with the row-wise normalization in Proposition 2, it is one of two mechanisms by which the deployed decision rule departs from an independent per-label classifier.*

Proof. Expanding $\mathbf{pM}'$ gives $\mathbf{p}((1-\lambda)\mathbf{I}+\lambda\mathbf{M})$, so the l-th component is $(1-\lambda)p_l + \lambda\sum_k p_k M_{kl}$. If $\mathbf{M}'$ were diagonal, each adjusted label score would depend only on its own projection score p_l. The off-diagonal entries M_{kl} for $k \neq l$ therefore introduce explicit cross-label coupling through co-occurrence statistics.

Proposition 2 (Ranking invariance under row-wise normalization). *Let* $\mathbf{u}_i = \mathbf{p}_i\mathbf{M}'$ *be the pre-normalization score vector from Proposition 1, and let* $\mathbf{s}_i = \text{normalize}(\mathbf{u}_i)$ *(row-wise min–max into* $[0,1]$*) be the score vector actually thresholded by the implementation. For any two labels* k, l *evaluated on the same instance* i*,*

$$s_{i,k} \geq s_{i,l} \iff u_{i,k} \geq u_{i,l};$$

in particular, at $\lambda = 0$ *(so* $u_{i,l} = p_{i,l} = \langle l \mid r_i\rangle^2$*), this ranking equals the ranking by cosine similarity* $\langle l \mid r_i\rangle$*. However, for a fixed global threshold* $\tau \in (0,1)$*, the decision* $s_{i,l} \geq \tau$ *is equivalent to*

$$u_{i,l} \;\geq\; \min_k u_{i,k} + \tau\Big(\max_k u_{i,k} - \min_k u_{i,k}\Big) =: \tau_i,$$

where τ_i *depends on instance* i *through the scores of* every *label on that instance, not only label* l*. Consequently,* τ *does not correspond to a single instance-independent cutoff on* $u_{i,l}$ *(or, at* $\lambda = 0$*, on cosine similarity): two instances with an identical raw score for label* l *can receive opposite decisions for that label if their scores on other labels differ.*

Proof. For fixed i with $\max_k u_{i,k} > \min_k u_{i,k}$, $s_{i,l} = (u_{i,l} - \min_k u_{i,k})/(\max_k u_{i,k} - \min_k u_{i,k})$ is a strictly increasing affine function of $u_{i,l}$ with the same slope and intercept for every label l of instance i, which proves the ranking-invariance claim. At $\lambda = 0$, $u_{i,l} = \langle l \mid r_i\rangle^2$, and since $\langle l \mid r_i\rangle = \cos\theta_{r_i,l} \geq 0$ (TF-IDF features and positive centroids are non-negative), squaring is order-preserving on $[0,\infty)$, so the ranking by $u_{i,l}$ equals the ranking by $\cos\theta_{r_i,l}$. Solving $s_{i,l} \geq \tau$ for $u_{i,l}$ using the definition of $s_{i,l}$ gives the threshold τ_i above, which is a function of $\min_k u_{i,k}$ and

$\max_k u_{i,k}$ and therefore of the scores of labels $k \neq l$ on instance i; two instances with equal $u_{i,l}$ but different label-score profiles can therefore yield different values of τ_i relative to $u_{i,l}$, hence different decisions. When $\max_k u_{i,k} = \min_k u_{i,k}$, the implementation sets $s_{i,l} = 0$ for all l, so no label is predicted for instance i at any $\tau > 0$.

Remark 1. Concretely, with two labels and $\tau = 0.5$: an instance with $\mathbf{u} = (0.5, 0.0)$ has $s_1 = (0.5 - 0)/(0.5 - 0) = 1.0 \geq \tau$, so label 1 is predicted; an instance with $\mathbf{u} = (0.5, 0.9)$ has the same raw score $u_1 = 0.5$ but $s_1 = (0.5 - 0.5)/(0.9 - 0.5) = 0 < \tau$, so label 1 is not predicted. The row-wise normalization is therefore a second source of cross-label coupling, independent of and in addition to the interference matrix $\mathbf{M}'$: even at $\lambda = 0$, with interference fully switched off, the model's per-label decisions are not those of L independent per-label centroid classifiers sharing one global threshold.

5 Experimental Setup

5.1 Datasets

NICE multi-label dataset. The main experiment uses the NICE dataset from Zenodo [13]. The downloaded CSV contains 622 requirements and binary label columns for NFR classes. After filtering rows with at least one NFR sub-class label and removing the empty "Other" label, the experiment uses 381 requirements and 11 NFR labels. Among these, 125 requirements contain more than one NFR label, giving a label cardinality of 1.3648.

PROMISE-expanded dataset. A secondary experiment uses PROMISE-expanded from the public requirements classification repository [11]. This dataset is treated as an 11-class single-label NFR subtype classification task and is used as an auxiliary sanity check rather than the main multi-label experiment.

5.2 Compared Models

The compared baselines are:

- **Label-frequency baseline:** predicts labels according to training label frequencies.
- **TF-IDF Logistic Regression:** one-vs-rest Logistic Regression using unigram and bigram TF-IDF features.
- **TF-IDF Linear SVM:** one-vs-rest Linear SVM using unigram and bigram TF-IDF features.
- **Sentence-BERT + LR:** all-MiniLM-L6-v2 embeddings followed by one-vs-rest Logistic Regression.

The quantum-inspired models are:

- **Quantum-inspired Projection:** positive-centroid semantic projection with label interference.

- **Contrastive Quantum Projection:** positive-minus-negative semantic projection with label interference.
- **Hybrid Quantum–SVM Fusion:** score-level fusion of contrastive quantum projection and Linear SVM, using a quantum fusion weight of 0.30.

5.3 Metrics and Threshold Selection

The main reported metrics are Micro-F1, Macro-F1, Hamming loss, and Label Ranking Average Precision (LRAP). Macro-F1 is emphasized because the NFR labels are imbalanced. Two threshold protocols are evaluated. First, a single global threshold is selected on an internal validation split by maximizing Macro-F1 over thresholds from 0.05 to 0.95. Second, a per-label threshold protocol selects an independent threshold for each NFR label on the validation split.

5.4 Hyperparameters

Table 1 summarizes the main hyperparameters used for reproducibility.

Table 1. Main hyperparameters used in the experiments.

Component	Setting
TF-IDF features	unigrams+bigrams, min_df=1
Sublinear TF-IDF	enabled for contrastive, hybrid, and SVM-only ablation
Logistic Regression	one-vs-rest, class_weight=balanced, max_iter=3000
Linear SVM	one-vs-rest LinearSVC, class_weight=balanced
Sentence-BERT	all-MiniLM-L6-v2 + one-vs-rest Logistic Regression
Positive projection interference	$\lambda = 0.15$
Contrastive projection interference	$\lambda = 0.05$
Hybrid projection interference	$\lambda = 0.03$
Hybrid quantum weight	$\alpha = 0.30$ unless varied in ablation
Threshold grid	0.05 to 0.95 with step 0.05
Random seed	42

6 Results

6.1 Single Train/Validation/Test Split on NICE

Table 2 reports results from a single train/validation/test split on the NICE dataset.

On this single split, Sentence-BERT obtains the highest Macro-F1 and LRAP. The positive projection variant performs better than the rectified contrastive variant, while the hybrid quantum–SVM model and the contrastive quantum projection share the lowest Hamming loss on this split. Because the split contains only 381 usable instances, these single-split results should be interpreted cautiously.

Table 2. NICE multi-label results on a single train/validation/test split, comparing label-frequency, TF-IDF, Sentence-BERT, and quantum-inspired models across Micro-F1, Macro-F1, Hamming loss, and LRAP. Sentence-BERT has the highest Macro-F1 and LRAP on this split; the hybrid quantum–SVM model and the contrastive quantum projection share the lowest Hamming loss.

Model	Threshold	Micro-F1	Macro-F1	Hamming Loss	LRAP
Label-frequency baseline	0.05	0.2221	0.2166	0.8751	0.4183
TF-IDF Logistic Regression	0.45	0.6299	0.5266	0.0901	0.7861
TF-IDF Linear SVM	0.45	0.6212	0.5205	0.0877	0.7843
Sentence-BERT + LR	0.50	**0.6667**	**0.6049**	0.0901	**0.8286**
Quantum-inspired Projection	0.85	0.6076	0.5618	0.0980	0.7694
Contrastive Quantum Projection	0.50	0.4907	0.3885	**0.0870**	0.6818
Hybrid Quantum–SVM Fusion	0.45	0.6284	0.5286	**0.0870**	0.7779

6.2 5-Fold Cross-validation on NICE

Table 3 reports 5-fold cross-validation results with mean and standard deviation across folds.

Table 3. Mean and standard deviation of 5-fold cross-validation results on the NICE multi-label dataset. Sentence-BERT achieves the highest LRAP; the hybrid quantum–SVM model obtains the highest Micro-F1 and Macro-F1 among non-embedding models, while TF-IDF Logistic Regression obtains the lowest Hamming loss.

Model	Micro-F1	Macro-F1	Hamming Loss	LRAP
Sentence-BERT + LR	0.6493 ± 0.0357	**0.6121 ± 0.0450**	0.0919 ± 0.0200	**0.8361 ± 0.0225**
Hybrid Quantum–SVM Fusion	**0.6839 ± 0.0331**	0.6084 ± 0.0294	0.0768 ± 0.0074	0.8040 ± 0.0266
TF-IDF Linear SVM	0.6821 ± 0.0405	0.6049 ± 0.0336	0.0764 ± 0.0094	0.8056 ± 0.0262
TF-IDF Logistic Regression	0.6787 ± 0.0463	0.5977 ± 0.0229	**0.0759 ± 0.0140**	0.7990 ± 0.0240
Contrastive Quantum Projection	0.4300 ± 0.0783	0.4118 ± 0.0275	0.2074 ± 0.1634	0.6968 ± 0.0503
Quantum-inspired Projection	0.5931 ± 0.0166	0.5490 ± 0.0237	0.1096 ± 0.0213	0.7830 ± 0.0304
Label-frequency baseline	0.2328 ± 0.0034	0.2083 ± 0.0032	0.7927 ± 0.0019	0.4059 ± 0.0110

Under standard 5-fold cross-validation with a global threshold, Sentence-BERT is the strongest Macro-F1 and LRAP baseline. The hybrid quantum–SVM fusion model is the strongest non-embedding model in Micro-F1 and Macro-F1 and improves over both pure quantum-inspired variants. A Wilcoxon signed-rank test over fold-level Macro-F1 finds no significant difference between the hybrid model and TF-IDF Linear SVM ($p = 0.1875$) or TF-IDF Logistic Regression ($p = 0.4375$). Given the small number of folds and the dependence induced by cross-validation resampling, these tests should be interpreted as descriptive evidence rather than a strong statistical conclusion [3].

6.3 5-Fold Cross-validation with Per-label Thresholds

Table 4 reports the 5-fold results when each label receives its own threshold selected on the validation split.

Table 4. Mean and standard deviation of 5-fold results with per-label threshold calibration on the NICE multi-label dataset. Sentence-BERT has the highest Macro-F1 and LRAP, while the hybrid quantum–SVM model remains the strongest non-embedding model in Macro-F1.

Model	Micro-F1	Macro-F1	Hamming Loss	LRAP
Sentence-BERT + LR	**0.6386 ± 0.0280**	**0.5990 ± 0.0276**	**0.0947 ± 0.0092**	**0.8361 ± 0.0225**
Hybrid Quantum–SVM Fusion	0.6178 ± 0.0749	0.5854 ± 0.0341	0.1027 ± 0.0476	0.8040 ± 0.0266
TF-IDF Linear SVM	0.6085 ± 0.0588	0.5640 ± 0.0309	0.1057 ± 0.0378	0.8056 ± 0.0262
TF-IDF Logistic Regression	0.6219 ± 0.0588	0.5629 ± 0.0532	0.0967 ± 0.0195	0.7990 ± 0.0240

With per-label threshold calibration, Sentence-BERT again obtains the best Macro-F1, and the hybrid quantum–SVM model remains the strongest non-embedding model in Macro-F1. Absolute Macro-F1 decreases for every model relative to the global-threshold protocol (Table 3), likely reflecting the smaller per-label validation subsets used to select each threshold; the hybrid model's relative drop (-3.8%) is smaller than TF-IDF Linear SVM's (-6.8%) or TF-IDF Logistic Regression's (-5.8%), suggesting the hybrid model is comparatively more robust to per-label threshold calibration rather than benefiting from it in absolute terms.

6.4 Ablation Study

Table 5 reports a 5-fold ablation study of the main quantum-inspired components. The ablation compares the original positive projection, contrastive projection with and without interference, an SVM-only component using the same sublinear TF-IDF representation as the hybrid model, and hybrid fusion with different quantum weights.

The ablation supports four observations. First, interference provides no measurable benefit for either projection variant on this dataset: it leaves the positive projection essentially unchanged (Macro-F1 0.5526 without interference versus 0.5490 with it) and substantially worsens the contrastive projection (0.5397 without versus 0.4118 with). The best hybrid configuration also uses a small interference weight ($\lambda = 0.03$), and the fusion weight that matters is α, not λ. We therefore treat interference as a secondary, largely inactive component in the current TF-IDF setting rather than as a primary contribution, and report it mainly because the underlying label-coupling formulation in Proposition 1 is independently interesting. Second, the rectified contrastive projection is semantically safer than squaring signed negative evidence, but in this dataset the pure contrastive variant performs worse than the positive projection. Third, the hybrid model improves over pure projection variants, showing the value of discriminative

Table 5. Ablation study on the NICE multi-label dataset. Hybrid fusion with $\alpha \in \{0.15, 0.30, 0.50\}$ performs within one standard deviation across all four metrics, indicating that the projection score is useful as a calibrated auxiliary signal but that Macro-F1 is not sensitive to the exact quantum weight in this range; $\alpha = 0.30$ is reported as the main configuration.

Variant	Micro-F1	Macro-F1	Hamming Loss	LRAP
Hybrid $\alpha = 0.15$	**0.6860 ± 0.0312**	**0.6091 ± 0.0293**	**0.0756 ± 0.0072**	**0.8085 ± 0.0280**
Hybrid $\alpha = 0.30$	0.6839 ± 0.0331	0.6084 ± 0.0294	0.0768 ± 0.0074	0.8040 ± 0.0266
Hybrid $\alpha = 0.50$	0.6814 ± 0.0404	0.6054 ± 0.0299	0.0795 ± 0.0113	0.8012 ± 0.0232
SVM-only sublinear TF-IDF	0.6713 ± 0.0553	0.5945 ± 0.0409	0.0843 ± 0.0263	0.8081 ± 0.0291
Positive projection without interference	0.5948 ± 0.0238	0.5526 ± 0.0285	0.1108 ± 0.0233	0.7839 ± 0.0309
Positive projection with interference	0.5931 ± 0.0166	0.5490 ± 0.0237	0.1096 ± 0.0213	0.7830 ± 0.0304
Contrastive projection without interference	0.5447 ± 0.0420	0.5397 ± 0.0398	0.1339 ± 0.0100	0.7183 ± 0.0523
Contrastive projection with interference	0.4300 ± 0.0783	0.4118 ± 0.0275	0.2074 ± 0.1634	0.6968 ± 0.0503

calibration. Compared with the SVM-only sublinear TF-IDF component, Hybrid $\alpha = 0.30$ improves mean Macro-F1 from 0.5945 to 0.6084, an absolute gain of 0.0139 attributable to adding the projection score as an auxiliary signal. Fourth, performance is best when the projection score is used as an auxiliary signal rather than as the dominant score.

6.5 Explainability Evaluation

We evaluate explanation faithfulness using the ERASER framework metrics: Comprehensiveness and Sufficiency. Across 158 true label assignments in the test split, for the contrastive projection model: (i) Comprehensiveness (score drop when deleting top-3 terms) was 0.0305, which is 5.21 times higher than random deletion (0.0059). (ii) Sufficiency (score change when keeping only top-3 terms) was -0.0395, whereas keeping 3 random terms dropped the score by 0.0337 (a sufficiency ratio of -1.17 compared to random). The negative sufficiency indicates that removing the non-rationale tokens slightly increases the projection score, filtering out background noise. For the TF-IDF Linear SVM baseline: (i) Comprehensiveness was 0.0867 (4.01 times random drop of 0.0216). (ii) Sufficiency was 0.0484 (0.42 times random drop of 0.1141). Relative to a random-deletion control, the intrinsic quantum-inspired explanations are meaningfully faithful and of a similar order to the SVM coefficient baseline; in absolute score-drop terms, the SVM explanation removes more score mass (0.0867 vs. 0.0305), so this comparison should be read as competitive on a relative, random-normalized basis rather than as uniformly stronger.

Table 6 gives a qualitative example from the same deletion test. The highest-contribution terms for the Security label correspond to access control concepts that are also plausible human rationales.

6.6 Auxiliary PROMISE-expanded Result

The auxiliary single-label PROMISE-expanded experiment suggests that the quantum-inspired projection model can be competitive in a single-label setting:

Table 6. Example requirement with top token-level contributions for the true NFR label, illustrating the intrinsic explanation produced by the projection model.

Requirement excerpt	Label	Top terms
"The product shall ensure that it can only be accessed by authorized users"	Security	authorized, only, access

on one 70/30 train/test split, it obtains 0.7025 accuracy and 0.6748 Macro-F1, compared with 0.6899 accuracy and 0.6344 Macro-F1 for TF-IDF Logistic Regression. This result comes from a single split without cross-validation or a significance test, so, like the single-split NICE result above, it should be interpreted cautiously; it is not the main evidence for the multi-label research claim, and because PROMISE-expanded is single-label, it does not exercise the multi-label interference mechanism.

7 Discussion

The experiments provide a balanced picture. On the main NICE multi-label dataset, Sentence-BERT is the strongest baseline in Macro-F1 and LRAP, consistent with the expectation that semantic embeddings are strong baselines for text classification. As a standalone classifier, the quantum-inspired projection is significantly weaker than TF-IDF Linear SVM: both the positive-centroid and contrastive variants lose to the SVM baseline in all 5 cross-validation folds, with large paired effect sizes (Cohen's $d_z = -1.36$ and -5.92, respectively; Wilcoxon $p = 0.0625$, the smallest value attainable at $n = 5$ folds). The quantum-inspired projection is therefore not, by itself, a competitive classifier.

The rectified contrastive quantum variant avoids rewarding evidence in the opposite label direction, but its pure projection performance is weaker than the positive-centroid projection on this dataset. Fusing the quantum score with a linear SVM changes this picture only partially: the hybrid model's Macro-F1 is directionally higher than an SVM-only baseline sharing the same TF-IDF representation (+0.0139, paired $d_z = 0.49$), but the isolated per-fold comparison shows this gain is fragile rather than robust — it is negative in 2 of 5 folds, and the positive mean is driven largely by a single fold. Across the main comparisons in Tables 3 and 4, every Wilcoxon test ($p \geq 0.1875$) is non-significant. Reliably detecting an effect of this size ($d_z \approx 0.5$) would require on the order of 30–35 paired cross-validation folds at conventional power, which is not attainable from one 381-instance dataset under 5-fold cross-validation; the null result reflects limited statistical power as much as a small effect. We therefore report the hybrid model's accuracy advantage over classical baselines as suggestive rather than confirmed.

To reduce reliance on fold-level significance testing, we also ran a paired bootstrap over the held-out NICE test requirements. The hybrid model's Macro-F1 advantage over TF-IDF Linear SVM was 0.0081 with a 95% bootstrap interval of [-0.0100, 0.0263], while its advantage over TF-IDF Logistic Regression was 0.0020 with interval [-0.0205, 0.0227]. Both intervals overlap zero, corroborating the cross-validation result: at the current dataset size, the data cannot distinguish the hybrid model's accuracy from that of TF-IDF baselines. The one accuracy comparison that is both large and well-supported is not between the hybrid model and classical baselines, but between the hybrid model and the pure quantum-inspired projection it fuses (+0.140 Macro-F1, 95% CI $[0.074, 0.213]$, $p < 0.001$): the quantum projection is too weak to deploy on its own and depends on the SVM component to be usable, rather than the reverse.

Although Sentence-BERT obtains the highest Macro-F1 and LRAP, the quantum-inspired hybrid model occupies a different point in the accuracy–interpretability trade-off, without a confirmed accuracy advantage over classical baselines to offer alongside it. Sentence-BERT relies on a pretrained transformer encoder and produces dense embeddings whose individual dimensions are not directly interpretable, so explanations require post-hoc methods such as LIME or SHAP [15,10]. In contrast, the projection model yields token-level contributions that are intrinsic to the scoring function and pass a deletion-based faithfulness check, at accuracy that is statistically indistinguishable from TF-IDF baselines and requires no pretrained model. The method is therefore most attractive when intrinsic interpretability and a small dependency footprint matter more than maximum Macro-F1, not as an accuracy improvement in its own right.

8 Threats to Validity

Dataset size and availability. NICE contains only a few hundred NFR-labeled requirements after filtering, and every cross-validation and ablation result in this paper is computed from this single 381-instance sample. A single dataset cannot establish that the observed model ranking would replicate on a differently sourced NFR corpus, since NICE's specific vocabulary, label distribution, and annotation conventions could each independently drive part of the ranking. Multi-label NFR datasets with fine-grained requirement subclasses are scarce, and to the best of our knowledge NICE is the only publicly available dataset suitable for evaluating multi-label co-occurrence and interference at the time of writing. Verifying generalization would require a second multi-label NFR corpus drawn from a different requirements source, with its own label co-occurrence structure, so that the interference formulation and the hybrid fusion behavior could be checked against label correlations the model was not implicitly tuned to. As a partial mitigation, we report an auxiliary single-label sanity check on the PROMISE-expanded dataset (525 NFR requirements), where the projection model remained competitive with a TF-IDF Logistic Regression baseline on one split; because PROMISE-expanded is single-label, this exercises the projection component but not the multi-label interference mechanism, and does not substitute for a second multi-label dataset. We consider the current evidence sufficient for this paper's central claims, which concern interpretability and a diagnostic account of where the quantum-inspired geometry helps or does not, but explicitly insufficient to support any claim that the observed accuracy ranking generalizes beyond NICE.

Baseline scope. The current implementation includes TF-IDF baselines and a frozen Sentence-BERT encoder, but not a fine-tuned transformer (BERT, RoBERTa, or NoRBERT), ML-kNN, or the small-language-model classifier from the NICE paper. A fine-tuned transformer would very likely obtain the highest Macro-F1 of any baseline in this comparison, since fine-tuning adapts all encoder parameters to the NFR classification task rather than using a frozen general-purpose sentence embedding; its absence means the gap reported here between the quantum-inspired hybrid model and the strongest baseline (Sentence-BERT) is a lower bound on the true gap to the strongest feasible classical model, not an estimate of it. We leave this comparison to future work rather than report a single, non-cross-validated fine-tuning run, which would not meet the reproducibility standard — cross-validated, seed-controlled, and checked against a pinned dependency environment — that we impose on every other model in this study. This does not affect the paper's central claims, which compare the quantum-inspired hybrid model against classical TF-IDF baselines and a frozen embedding baseline, and which are explicitly framed around interpretability rather than establishing state-of-the-art accuracy.

Model simplicity. The quantum-inspired models are based on TF-IDF semantic states, projection directions, and score fusion. They do not yet use contextual neural embeddings inside the projection component.

Threshold sensitivity. Multi-label classification depends strongly on decision thresholds. The experiments show that per-label threshold calibration can change the relative ranking of models.

Explainability evaluation. The deletion test measures faithfulness to the model's own scoring function, but it does not measure human usefulness or agreement with expert rationales.

Ablation scope. The ablation study examines projection type, interference, and fusion weight, but it does not yet test contextual embeddings inside the quantum-inspired projection component.

9 Conclusion

This paper presented explainable quantum-inspired models for multi-label NFR classification. Requirements are represented as semantic states, NFR labels are modeled as projection directions, and predictions are explained through semantic amplitude contributions. As a standalone classifier, the quantum-inspired projection is significantly weaker than classical TF-IDF baselines (large effect size, all cross-validation folds); fused with a linear SVM, its Macro-F1 advantage over TF-IDF and Sentence-BERT baselines is small, directionally positive, and not statistically significant at the current dataset size. The paper's contribution is therefore an interpretability result and a diagnostic ablation — showing where the quantum-inspired geometry helps (as an auxiliary fusion signal) and where it does not (as a standalone classifier, and under label interference) — rather than a confirmed accuracy improvement. The deletion-based explainability evaluation shows that top contribution terms are substantially more influential than random terms, and this intrinsic explanation is obtained without a post-hoc method or a pretrained model. The ablation study indicates that hybrid fusion is the most useful quantum-inspired component in the current implementation, while pure contrastive projection requires further refinement.

Future work should integrate contextual embeddings into the quantum-inspired representation, compare with fine-tuned BERT/RoBERTa [9] and NoRBERT baselines under the same cross-validated protocol used here, evaluate ML-kNN [20] as a classical multi-label baseline, test the approach on a second multi-label NFR corpus to check generalization beyond NICE, and collect human rationale judgments for explanation quality.

Reproducibility

The implementation contains scripts for reproducing the experiments:

- `scripts/run_nice_multilabel_experiment.py`
- `scripts/run_nice_cv_experiment.py`
- `scripts/run_nice_per_label_threshold_experiment.py`
- `scripts/run_nice_ablation_experiment.py`
- `scripts/run_explainability_deletion_test.py`
- `scripts/run_nice_robustness_experiment.py`
- `scripts/run_all_experiments.py`
- `scripts/run_promise_experiment.py`

For anonymous review, the artifact is available at the anonymized repository: https://anonymous.4open.science/r/nfr-review-artifact-CE2D/.

Use of AI Tools

During the preparation of this manuscript, the authors used a large language model to assist with: (i) re-running and verifying the experimental pipeline for reproducibility, including diagnosing a dependency-pinning and random-seed issue that had made prior results non-reproducible; (ii) computing the effect-size, bootstrap, and significance analyses reported in the Discussion; (iii) deriving and verifying the corrected statement and proof of Proposition 2; and (iv) drafting and revising text throughout the manuscript, including the framing of the paper's contribution. All AI-assisted code, analysis, and text were reviewed, independently checked against the underlying experimental data, and verified by the authors, who take full responsibility for the content of the paper.

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